

# ■ AgriPredict — Complete Technical Architecture & Feature Guide

**Exhaustive Blueprint:** Deep-dive explanation of every module, feature, API integration, data model, and engineering choice in AgriPredict.

## 1. System Philosophy & Executive Purpose (Why it is Used)

**The Core Agricultural Problem in India:** Over 140 million farming households operate across 28 States and 8 Union Territories. However, Indian agriculture suffers from extreme information asymmetry. Local traders, middlemen, and commission agents frequently buy crops at artificially depressed farm-gate prices. Farmers lack direct, real-time access to nearby APMC mandi prices, retail market rates, localized weather risk forecasts, and crop-specific demand trajectories.

**How AgriPredict Solves This:** AgriPredict is a zero-cost, high-speed AI market intelligence platform. It provides Indian farmers with direct price transparency across 3,000+ APMC mandis, identifies the highest profit market for their crop, delivers 30-day AI price forecasting, decodes complex government schemes, and provides a 6-language voice-activated AI assistant.

## 2. Deep Technology Stack & Architecture (What We Used & How)

| Layer              | Technologies Used                               | Technical Function & Engineering Rationale                                                                                                     |
|--------------------|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Frontend Core      | Next.js 16 (App Router)<br>React 19, TypeScript | Server-Side Rendering (SSR) + Client State. Turbopack compiler ensures instant builds. TypeScript guarantees 100% type-safe data pipelines.    |
| Design & Styling   | Tailwind CSS<br>Custom CSS Tokens               | Warm Earthy palette (--primary: #5C7A3E). Responsive grid system scaling smoothly from 320px mobile to 1200px+ desktop frames.                 |
| AI Core Engine     | Google Gemini 1.5 Flash API                     | Used for real-time market intelligence synthesis, 30-day price forecasting, crop trade recommendation, and document OCR translation.           |
| APMC Market Data   | Agmarknet Open Data API<br>(`api.data.gov.in`)  | Official Govt of India Mandi API. Returns Min, Max, Modal Prices (₹/q & ₹/kg) and arrival volumes for 3,000+ mandis.                           |
| Weather Radar      | Open-Meteo Weather API                          | No-key REST API returning live temperature (°C), humidity (%), wind speed, WMO codes, and 7-day soil moisture indices.                         |
| Geolocation        | OpenStreetMap Nominatim<br>+ BigDataCloud API   | Dual-provider reverse geocoding engine. Automatically converts browser GPS coordinates into Indian State & District.                           |
| Voice & Speech     | HTML5 Web Speech API                            | Voice recognition in 6 Indian languages + Roman transliterated scripts ('Tanglish', 'Teluglish', 'Hinglish'). Speech Synthesis response audio. |
| Document Scanner   | HTML5 Canvas + Gemini Vision API                | Renders uploaded PDFs/images and extracts structured plain-language summaries from PM-KISAN, KCC, and Soil cards.                              |
| Data Visualization | Recharts Library                                | SVG line charts for 12-month historical price curves, monthly volatility indices, and trade volume metrics with dark mode support.             |

## 3. Exhaustive Feature-by-Feature Engineering Breakdown

**Feature 1: Real-Time APMC Mandi Price Comparison (`src/components/MandiComparison.tsx`)**

- **How it is Used:** Farmers select State, District, Crop, and Variety from dropdowns to view live mandi rates in their area.
- **Why it is Used:** Enables farmers to compare wholesale rates across nearby markets and choose the highest-paying market.
- **How it was Made:** Built using a dedicated dataset (`STRICT\_DISTRICT\_MANDIS` in mandiDatabase.ts) covering 28 States, 8 UTs, 780+ Districts, and 3,000+ Mandis (10+ mandis per district). Connected to `/api/mandi-prices` API route.
- **Key Highlights:** Displays Wholesale Rates (■/quintal & ■/kg), Retail Rates, Arrival Volumes (Tons/day), Trend (Rising/Stable/Falling), and highlights the ■ **Highest Price Mandi**.

**Feature 2: Crop Varieties & Cultivar Intelligence Grid (`src/app/crops/page.tsx`, `src/lib/cropData.ts`)**

- **How it is Used:** View and filter specific varieties for 28 major crops (e.g. Alphonso vs Totapuri Mangoes, Grand Naine Banana, Basmati 1121 Rice, Guntur S4 Chilli).
- **Why it is Used:** Prices differ drastically depending on crop variety (e.g. Alphonso ■65/kg mandi vs Totapuri ■8/kg mandi).
- **How it was Made:** Created `CropVariety` interface in `cropData.ts` with `mandiPricePerKg` & `pricePerKg` fields, rendered in an interactive 3-column responsive card grid.

**Feature 3: AI Price Recommendation & 30-Day Market Forecast Engine**

- **How it is Used:** Displays prominent **BUY**, **SELL**, or **HOLD** status badges, projected 30-day percentage trend (e.g., ↑ +5.2%), confidence score (%), and crop risk rating.
- **Why it is Used:** Prevents panic selling during temporary price drops and helps farmers time their harvest for maximum profit.
- **How it was Made:** Uses `generatePriceHistory()`, `generateForecast()`, and `getRecommendation()` logic combining moving averages, supply arrivals, and weather impacts.

**Feature 4: Multilingual Voice AI Chatbot Assistant (`src/components/ChatbotVoice.tsx`, `/chatbot`)**

- **How it is Used:** Farmers click the microphone icon to speak questions or type in text across 6 supported languages (English, ■■■■■■, ■■■■■■, ■■■■■■, ■■■■■■, ■■■■■■).
- **Why it is Used:** Allows non-English speaking and illiterate farmers to query prices, pest controls, and schemes effortlessly in their native tongue.
- **How it was Made:** Powered by Gemini 1.5 Flash via `/api/chatbot/route.ts`. Uses browser `webkitSpeechRecognition` with language tags (`ta-IN`, `te-IN`, `hi-IN`, etc.) and `window.speechSynthesis` audio playback.

**Feature 5: Smart Government Document OCR Scanner (`src/components/DocumentOcr.tsx`, `/ocr`)**

- **How it is Used:** Farmers upload an image or scan of any government document (PM-KISAN notice, Kisan Credit Card loan sanction, Soil Card).
- **Why it is Used:** Government circulars contain complex legal jargon. Farmers get structured, plain-language summaries and 3-4 actionable next steps in 6 languages.
- **How it was Made:** Built with HTML5 Canvas API + Gemini Multimodal Vision API via `/api/ocr/route.ts` returning strict JSON structured outputs.

**Feature 6: Government Welfare Schemes Portal (`src/app/schemes/page.tsx`)**

- **How it is Used:** Search and filter major schemes (PM-KISAN, PMFBY, KCC, PMKSY Drip Irrigation, Agri Infrastructure Fund), run eligibility checks, and apply directly.
- **Why it is Used:** Millions of Indian farmers lose out on government subsidies due to lack of awareness.
- **How it was Made:** Structured `SCHEMES` dataset in `schemeData.ts` with category filters, eligibility matchers, modal views, and official government portal links.

**Feature 7: Market Analytics & Volatility Index Dashboard (`src/app/analytics/page.tsx`)**

- **How it is Used:** View 12-month historical price curves, monthly volatility indices, and trade volume metrics.

- **Why it is Used:** Helps agronomists, traders, and farmers spot long-term seasonal market cycles.
- **How it was Made:** Interactive SVG charts powered by Recharts library with custom tooltips, dark mode axes, and multi-crop comparison toggles.

### Feature 8: My Crops Personal Farm Portfolio (`src/app/my-crops/page.tsx`)

- **How it is Used:** Farmers bookmark their grown crops, enter farm acreage, track total projected revenue, and set custom price alert thresholds.
- **Why it is Used:** Creates a personalized operational dashboard tailored to the farmer's specific land size and crops.
- **How it was Made:** LocalStorage state management (`agri\_portfolio`), dynamic revenue formula (`acreage \* yield \* price`), and state-persistent crop cards.

### Feature 9: Live Weather Radar & Soil Moisture Radar (`src/lib/weather.ts`)

- **How it is Used:** View real-time temperature, condition icons, relative humidity (%), wind speed (km/h), 7-day rainfall, and soil moisture index (%).
- **Why it is Used:** Weather and soil moisture dictate ideal harvesting windows and disease risk.
- **How it was Made:** REST API integration with Open-Meteo (`api.open-meteo.com`) with WMO code dictionary and `getCropImpact()` risk evaluation.

### Feature 10: Automatic Live GPS Location Detection (`src/lib/AppContext.tsx`)

- **How it is Used:** Click `■ Use Live Location` in sidebar footer or settings page.
- **Why it is Used:** Automatically updates State, District, Weather, and APMC Mandi prices to the farmer's exact location without typing.
- **How it was Made:** Uses `navigator.geolocation` + OpenStreetMap Nominatim / BigDataCloud reverse geocoding APIs + Krishnankovil/Madurai locality alias mapping + 2-way `AppContext` sync.

## 4. Summary & Impact Statement

**Summary:** AgriPredict integrates modern Next.js 16 web technology, Google Gemini AI, official Agmarknet government data, and Open-Meteo weather radar into a seamless, 0ms latency progressive web application. By making real-time APMC mandi prices accessible in 6 Indian languages with voice input, AgriPredict puts market power back into the hands of Indian farmers.